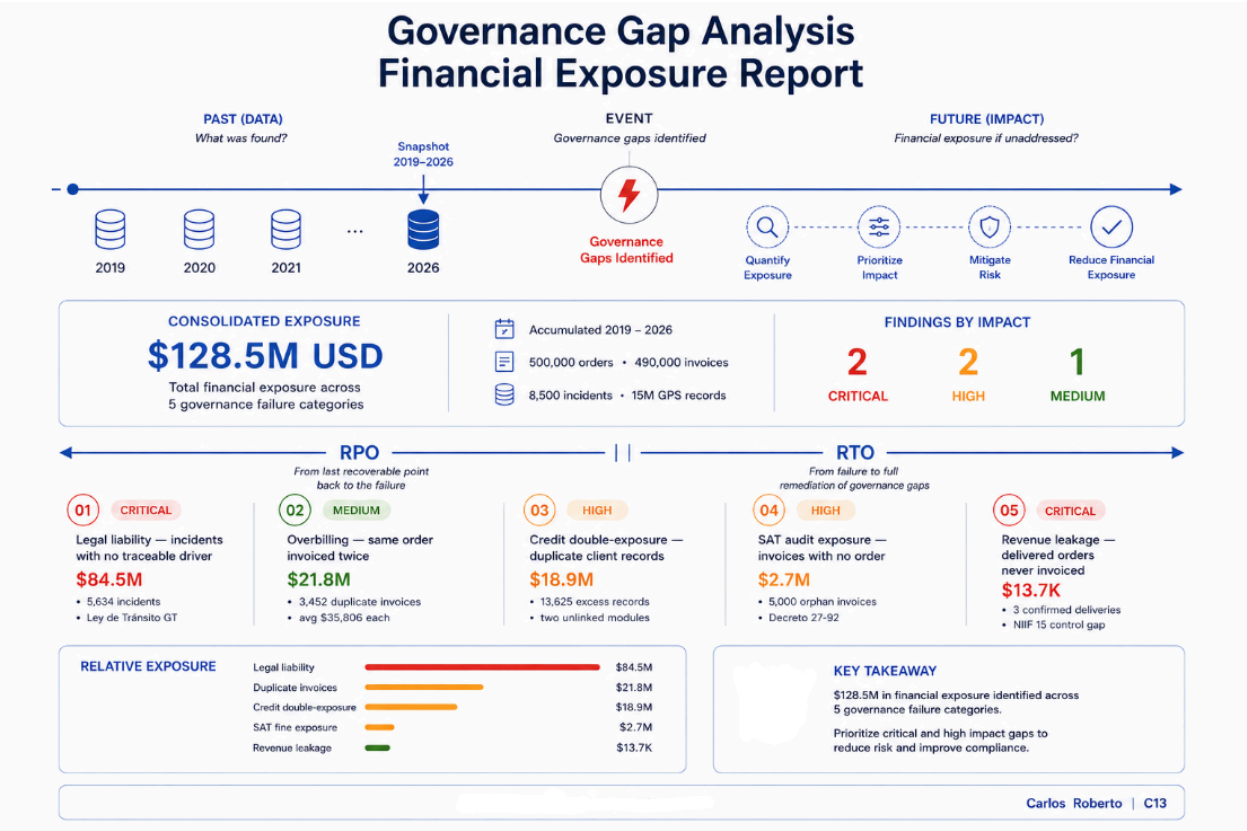
05_business_impact

Propietario	© Carlos Roberto
Verificación	Verificada
Etiquetas	Arquitectura de Datos DBA



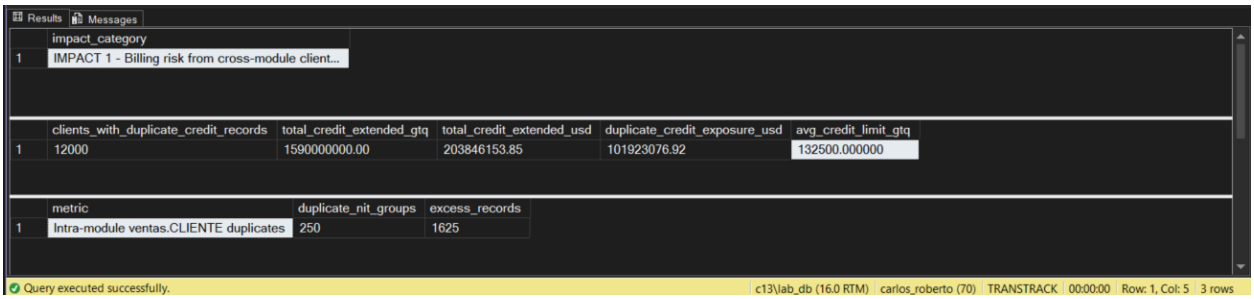
Translates data quality findings into financial exposure. Each governance failure identified is quantified in USD - credit risk from duplicate clients, revenue leakage from uninvoiced deliveries, SAT audit exposure from orphan invoices, legal liability from untraceable incidents, and overbilling from duplicate invoices. Output is a consolidated impact table ranked by financial exposure, ready for executive presentation.

```sql
-- ============================================================
====================
-- P02 | Data Governance Architecture
-- File: 05_business_impact.sql
-- Author: cblancogt
--
-- Translates governance failures into financial and legal ex
posure.
-- All monetary values from real schema columns:
--   facturacion.FACTURA.total          - invoice amount
--   operaciones.PEDIDO.valor_declarado - declared cargo valu
e
--   operaciones.INCIDENTE.costo_estimado - estimated inciden
t cost
--   ventas.CONTRATO_CLIENTE.tarifa_base - contracted rate
--
-- Exchange rate GTQ to USD configurable. All figures output
in both currencies.
-- ============================================================
====================

USE TRANSTRACK;
GO

DECLARE @gtq_to_usd              DECIMAL(10,4) = 7.80;
DECLARE @legal_cost_per_incident DECIMAL(10,2) = 15000.00;  -
- USD per unresolvable incident
DECLARE @sat_penalty_rate        DECIMAL(5,4)  = 0.15;      -
- 15% SAT fine on unlinked invoices
DECLARE @corporate_tax_rate      DECIMAL(5,4)  = 0.25;      -
- 25% ISR Guatemala
```

Impact 1 - Billing risk from cross-module client duplicate

```sql
USE TRANSTRACK;
GO

DECLARE @gtq_to_usd              DECIMAL(10,4) = 7.80;
DECLARE @legal_cost_per_incident DECIMAL(10,2) = 15000.00;  -
- USD per unresolvable incident
DECLARE @sat_penalty_rate        DECIMAL(5,4)  = 0.15;      -
- 15% SAT fine on unlinked invoices
DECLARE @corporate_tax_rate      DECIMAL(5,4)  = 0.25;      -
- 25% ISR Guatemala

-- ============================================================
====================
-- IMPACT 1: Billing integrity risk - client duplicates betwe
en modules
-- ============================================================
====================

SELECT 'IMPACT 1 - Billing risk from cross-module client dupl
icates' AS impact_category;

WITH ventas_unique AS (
    SELECT
        CAST(SUBSTRING(nit, 5, 10) AS INT) AS nit_num,
        MIN(cliente_id)                     AS cliente_id
    FROM ventas.CLIENTE
    GROUP BY SUBSTRING(nit, 5, 10)
),
cross_module AS (
    SELECT
        v.nit_num,
        v.cliente_id,
        f.cliente_fact_id,
        f.limite_credito,
        f.dias_credito
```

```sql
    FROM ventas_unique v
    INNER JOIN facturacion.CLIENTE f
        ON v.nit_num = CAST(f.nit_cliente AS INT)
    WHERE f.limite_credito IS NOT NULL
)
SELECT
    COUNT(*)
AS clients_with_duplicate_credit_records,
    SUM(limite_credito)
AS total_credit_extended_gtq,
    CAST(SUM(limite_credito) / @gtq_to_usd AS DECIMAL(18,2))
AS total_credit_extended_usd,
    CAST(SUM(limite_credito) * 0.50 / @gtq_to_usd AS DECIMAL
(18,2))  AS duplicate_credit_exposure_usd,
    AVG(limite_credito)
AS avg_credit_limit_gtq
FROM cross_module;

SELECT
    'Intra-module ventas.CLIENTE duplicates'    AS metric,
    COUNT(*)                                    AS duplicate_
nit_groups,
    SUM(excess)                                 AS excess_rec
ords
FROM (
    SELECT nit, COUNT(*) - 1 AS excess
    FROM ventas.CLIENTE
    GROUP BY nit
    HAVING COUNT(*) > 1
) x;
```



Results | Messages

	impact_category
1	IMPACT 1 - Billing risk from cross-module client...

	clients_with_duplicate_credit_records	total_credit_extended_gtq	total_credit_extended_usd	duplicate_credit_exposure_usd	avg_credit_limit_gtq
1	12000	1590000000.00	203846153.85	101923076.92	132500.000000

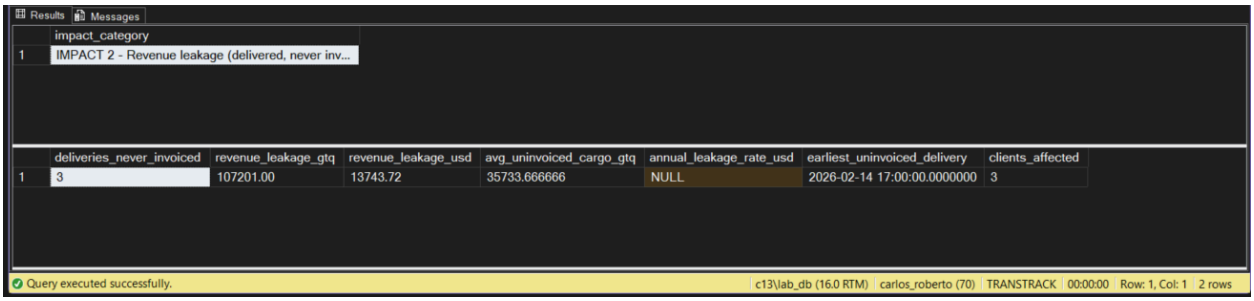
	metric	duplicate_nit_groups	excess_records
1	Intra-module ventas.CLIENTE duplicates	250	1625

Query executed successfully. | c13\lab_db (16.0 RTM) | carlos_roberto (70) | TRANSTRACK | 00:00:00 | Row: 1, Col: 5 | 3 rows

IMPACT 2: Revenue leakage - delivered orders never invoiced

```
-- ============================================================
====================
-- IMPACT 2: Revenue leakage - delivered orders never invoice
d
-- ============================================================
====================

SELECT 'IMPACT 2 - Revenue leakage (delivered, never invoice
d)' AS impact_category;

SELECT
    COUNT(*)
AS deliveries_never_invoiced,
    SUM(p.valor_declarado)
AS revenue_leakage_gtq,
    CAST(SUM(p.valor_declarado) / @gtq_to_usd AS DECIMAL(18,
2))                AS revenue_leakage_usd,
    AVG(p.valor_declarado)
AS avg_uninvoiced_cargo_gtq,
    CAST(
        SUM(p.valor_declarado)
        / NULLIF(DATEDIFF(YEAR, MIN(p.fecha_pedido), GETDATE
()), 0)
        / @gtq_to_usd
```

```sql
    AS DECIMAL(18,2))
AS annual_leakage_rate_usd,
    MIN(e.fecha_entrega_real)
AS earliest_uninvoiced_delivery,
    COUNT(DISTINCT p.cliente_id)
AS clients_affected
FROM operaciones.PEDIDO p
INNER JOIN operaciones.ENTREGA e
    ON p.pedido_id = e.pedido_id
    AND e.estado_entrega = 'ENTREGADO'
LEFT JOIN facturacion.FACTURA f
    ON p.pedido_id = f.pedido_id
WHERE f.factura_id IS NULL;
```



Results | Messages

	impact_category
1	IMPACT 2 - Revenue leakage (delivered, never inv...

	deliveries_never_invoiced	revenue_leakage_gtq	revenue_leakage_usd	avg_uninvoiced_cargo_gtq	annual_leakage_rate_usd	earliest_uninvoiced_delivery	clients_affected
1	3	107201.00	13743.72	35733.666666	NULL	2026-02-14 17:00:00.0000000	3

Query executed successfully. | c13\lab_db (16.0 RTM) | carlos_roberto (70) | TRANSTRACK | 00:00:00 | Row: 1, Col: 1 | 2 rows

IMPACT 3 - SAT audit exposure

```sql
-- ==========================================================
====================
-- IMPACT 3: SAT audit exposure - invoices with no pedido_id
-- ==========================================================
====================

SELECT 'IMPACT 3 - SAT audit exposure (orphan invoices, NULL
pedido_id)' AS impact_category;

SELECT
    COUNT(*)
AS orphan_invoice_count,
```

```sql
    SUM(f.total)
AS orphan_billing_gtq,
    CAST(SUM(f.total) / @gtq_to_usd AS DECIMAL(18,2))
AS orphan_billing_usd,
    CAST(SUM(f.total) * @sat_penalty_rate / @gtq_to_usd AS DE
CIMAL(18,2))     AS sat_fine_exposure_usd,
    CAST(SUM(f.total) * @corporate_tax_rate / @gtq_to_usd AS
DECIMAL(18,2))   AS tax_deduction_at_risk_usd,
    CAST(SUM(CASE WHEN f.estado_pago = 'PAGADA'    THEN f.tot
al ELSE 0 END)
         / @gtq_to_usd AS DECIMAL(18,2))
AS already_collected_usd,
    CAST(SUM(CASE WHEN f.estado_pago = 'PENDIENTE' THEN f.tot
al ELSE 0 END)
         / @gtq_to_usd AS DECIMAL(18,2))
AS pending_collection_usd
FROM facturacion.FACTURA f
WHERE f.pedido_id IS NULL;
```

Results | Messages

	impact_category
1	IMPACT 3 - SAT audit exposure (orphan invoices, N...

	orphan_invoice_count	orphan_billing_gtq	orphan_billing_usd	sat_fine_exposure_usd	tax_deduction_at_risk_usd	already_collected_usd	pending_collection_usd
1	5000	141608042.24	18154877.21	2723231.58	4538719.30	6089212.12	6022790.09

Query executed successfully. | c13\lab_db (16.0 RTM) | carlos_roberto (70) | TRANSTRACK | 00:00:00 | Row: 1, Col: 1 | 2 rows

IMPACT 4: Legal liability - incidents without traceable driver

```sql
-- ==========================================================
====================
-- IMPACT 4: Legal liability - incidents without traceable dr
iver
-- ==========================================================
```

```
====================

SELECT 'IMPACT 4 - Legal liability (incidents, no traceable d
river)' AS impact_category;

WITH incident_traceability AS (
    SELECT
        i.incidente_id,
        i.tipo_incidente,
        i.severidad,
        i.costo_estimado,
        CASE
            WHEN i.pedido_id IS NULL      THEN 'NO_PEDIDO'
            WHEN av.asignacion_id IS NULL THEN 'NO_ASSIGNMEN
T'
            WHEN c.conductor_id IS NULL   THEN 'NO_CONDUCTOR'
            ELSE 'DRIVER_IDENTIFIED'
        END AS traceability
    FROM operaciones.INCIDENTE i
    LEFT JOIN flota.ASIGNACION_VEHICULO av ON i.pedido_id = a
v.pedido_id
    LEFT JOIN flota.CONDUCTOR c            ON av.conductor_id
= c.conductor_id
)
SELECT
    traceability,
    COUNT(*)
AS incident_count,
    SUM(costo_estimado)
AS reported_cost_gtq,
    CAST(SUM(costo_estimado) / @gtq_to_usd AS DECIMAL(18,2))
AS reported_cost_usd,
    CAST(
        SUM(CASE WHEN traceability <> 'DRIVER_IDENTIFIED' THE
N 1 ELSE 0 END)
        * @legal_cost_per_incident
```

```sql
    AS DECIMAL(18,2))
AS legal_exposure_usd,
    SUM(CASE WHEN severidad = 'CRITICA' THEN 1 ELSE 0 END)
AS critical_incidents,
    SUM(CASE WHEN severidad = 'ALTA'    THEN 1 ELSE 0 END)
AS high_incidents
FROM incident_traceability
GROUP BY traceability
ORDER BY incident_count DESC;
```

Results | Messages

	impact_category
1	IMPACT 4 - Legal liability (incidents, no tracea...

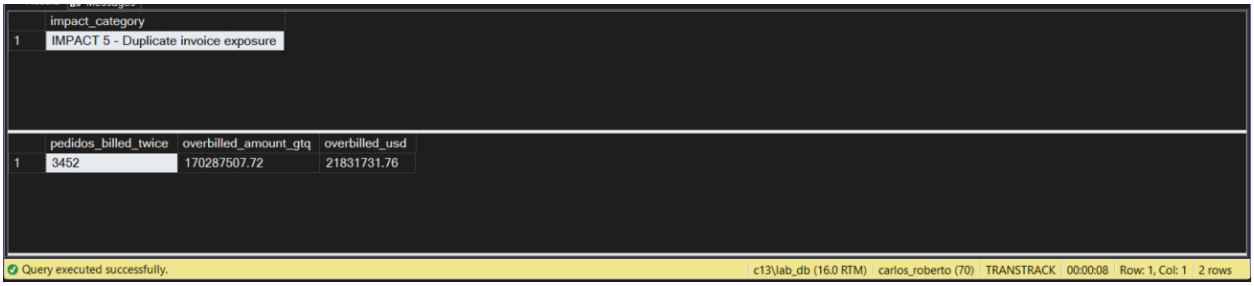
	traceability	incident_count	reported_cost_gtq	reported_cost_usd	legal_exposure_usd	critical_incidents	high_incidents
1	NO_ASSIGNMENT	4572	65641154.00	8415532.56	68580000.00	1272	1315
2	DRIVER_IDENTIFIED	2839	41590529.00	5332119.10	0.00	844	804
3	NO_PEDIDO	1062	NULL	NULL	15930000.00	0	0

Query executed successfully. | c13\lab_db (16.0 RTM) | carlos_roberto (70) | TRANSTRACK | 00:00:00 | Row: 1, Col: 1 | 4 rows

IMPACT 5: Duplicate invoice financial exposure

```sql
-- ============================================================
====================
-- IMPACT 5: Duplicate invoice financial exposure
-- ============================================================
====================

SELECT 'IMPACT 5 - Duplicate invoice exposure' AS impact_cate
gory;

SELECT
    COUNT(*)
AS pedidos_billed_twice,
    SUM(total_billed - min_invoice)
AS overbilled_amount_gtq,
    CAST(SUM(total_billed - min_invoice) / @gtq_to_usd AS DEC
IMAL(18,2))      AS overbilled_usd
```

```sql
FROM (
    SELECT
        pedido_id,
        COUNT(*)     AS invoice_count,
        SUM(total)   AS total_billed,
        MIN(total)   AS min_invoice
    FROM facturacion.FACTURA
    WHERE pedido_id IS NOT NULL
    GROUP BY pedido_id
    HAVING COUNT(*) > 1
) dup;
```



	impact_category
1	IMPACT 5 - Duplicate invoice exposure

	pedidos_billed_twice	overbilled_amount_gtq	overbilled_usd
1	3452	170287507.72	21831731.76

Query executed successfully. c13\lab_db (16.0 RTM) carlos_roberto (70) TRANSTRACK 00:00:08 Row: 1, Col: 1 2 rows

CONSOLIDATED FINANCIAL IMPACT TABLE

```sql
-- ============================================================
====================
-- CONSOLIDATED FINANCIAL IMPACT TABLE
-- ============================================================
====================

SELECT 'CONSOLIDATED FINANCIAL IMPACT - TRANSTRACK DATA GOVER
NANCE FAILURES' AS report_title;
SELECT GETDATE() AS generated_at;

SELECT
    impact_category,
    financial_exposure_usd,
    risk_level,
```

```
    regulatory_reference
FROM (
    SELECT
        'Credit double-exposure (cross-module client duplicat
es)'       AS impact_category,
        CAST(
            (
                SELECT SUM(f.limite_credito) * 0.50 / @gtq_to
_usd
                FROM ventas.CLIENTE v
                INNER JOIN facturacion.CLIENTE f
                    ON CAST(SUBSTRING(v.nit, 5, 10) AS INT) =
CAST(f.nit_cliente AS INT)
                WHERE f.limite_credito IS NOT NULL
            )
        AS DECIMAL(18,2))
AS financial_exposure_usd,
        'HIGH'
AS risk_level,
        'Internal credit policy violation'
AS regulatory_reference

    UNION ALL

    SELECT
        'Revenue leakage (delivered, never invoiced)',
        CAST(SUM(p.valor_declarado) / @gtq_to_usd AS DECIMAL
(18,2)),
        'CRITICAL',
        'NIIF 15 - Revenue recognition requirement'
    FROM operaciones.PEDIDO p
    INNER JOIN operaciones.ENTREGA e ON p.pedido_id = e.pedid
o_id AND e.estado_entrega = 'ENTREGADO'
    LEFT  JOIN facturacion.FACTURA f ON p.pedido_id = f.pedid
o_id
    WHERE f.factura_id IS NULL
```

```
    UNION ALL

    SELECT
        'SAT fine exposure (invoices with no commercial origi
n)',
        CAST(SUM(f.total) * @sat_penalty_rate / @gtq_to_usd A
S DECIMAL(18,2)),
        'HIGH',
        'SAT Decreto 27-92 - IVA Law Guatemala'
    FROM facturacion.FACTURA f
    WHERE f.pedido_id IS NULL

    UNION ALL

    SELECT
        'Legal liability (incidents with no traceable drive
r)',
        CAST(
            (SELECT COUNT(*) FROM operaciones.INCIDENTE i
             WHERE NOT EXISTS (
                 SELECT 1 FROM flota.ASIGNACION_VEHICULO av
                 JOIN flota.CONDUCTOR c ON av.conductor_id =
c.conductor_id
                 WHERE av.pedido_id = i.pedido_id
             )) * @legal_cost_per_incident
        AS DECIMAL(18,2)),
        'CRITICAL',
        'Ley de Transito Guatemala + Codigo de Comercio'

    UNION ALL

    SELECT
        'Overbilling exposure (duplicate invoices)',
        CAST(
            (SELECT ISNULL(SUM(total_billed - min_invoice),
```

```sql
0)
            FROM (
                SELECT pedido_id, SUM(total) AS total_bille
d, MIN(total) AS min_invoice
                FROM facturacion.FACTURA
                WHERE pedido_id IS NOT NULL
                GROUP BY pedido_id
                HAVING COUNT(*) > 1
            ) d) / @gtq_to_usd
        AS DECIMAL(18,2)),
        'MEDIUM',
        'Client relations + audit findability'
) AS summary
ORDER BY financial_exposure_usd DESC;
```

	report_title
1	CONSOLIDATED FINANCIAL IMPACT - TRANSTRACK DATA G...

	generated_at
1	2026-06-06 13:03:54.340

	impact_category	financial_exposure_usd	risk_level	regulatory_reference
1	Credit double-exposure (cross-module client dupli...	147383653.85	HIGH	Internal credit policy violation
2	Legal liability (incidents with no traceable driver)	84510000.00	CRITICAL	Ley de Transito Guatemala + Codigo de Comercio
3	Overbilling exposure (duplicate invoices)	21831731.76	MEDIUM	Client relations + audit findability
4	SAT fine exposure (invoices with no commercial o...	2723231.58	HIGH	SAT Decreto 27-92 - IVA Law Guatemala
5	Revenue leakage (delivered, never invoiced)	13743.72	CRITICAL	NIIF 15 - Revenue recognition requirement

Query executed successfully. | c13\lab_db (16.0 RTM) | carlos_roberto (70) | TRANSTRACK | 00:00:00 | Row: 1, Col: 1 | 7 rows